Week 2: Tables, forms and CSS

Tables

```
<table>
  <tr><th>Day</th><th>Subject</th></tr>
  <tr><td>Mon</td><td>Maths</td></tr>
  <tr><td colspan="2">Merged cell</td></tr>
</table>
```

- `<tr>` row, `<th>` heading cell, `<td>` data cell.
- `colspan="2"` merges cells to the right; `rowspan="2"` merges cells downwards. Count carefully: every row must add up to the same number of columns.
- Use tables for **data**, never for page layout.

Forms

```
<form action="/submit" method="get">
  <label for="name">Name</label>
  <input type="text" id="name" name="name" required>
  <button type="submit">Send</button>
</form>
```

Element	Notes
`<input type="text">`	One line of text. Also `email`, `password`, `number`, `date`
`<input type="radio">`	Choose one. Same `name`, different `value`
`<input type="checkbox">`	Tick any. Sends nothing if unticked
`<select>` `<option>`	Drop-down list
`<textarea>`	Several lines of text
`<button>`	Submits the form (`type="submit"`)

- `name` is the key the data is sent under. No `name` means the field is not sent.
- `for` + `id` join a label to its input: tapping the label focuses the field.
- `required`, `minlength`, `maxlength`, `min`, `max`, `pattern`, `placeholder` help the user, but are not security.

GET and POST

	GET	POST
Data goes	in the URL (`?name=Mei`)	in the request body
Use for	searching, looking	changing something
Bookmark?	yes	no

CSS: selecting and styling

```
p { color: #333; }           /* every <p> */
.card { background: #eef; }  /* class="card" */
#logo { width: 120px; }      /* id="logo" */
nav a:hover { color: red; }  /* hover state */
```

```
<!-- in the head: -->
<link rel="stylesheet" href="style.css">
```

Property	Effect
`color`, `background-color`	Text / background colour (`#1f4e8c`, `red`)
`font-family`, `font-size`	Font; `1rem` = the browser's base size (16px)
`text-align`	`left`, `center`, `right`
`margin` / `padding`	Space outside / inside the border
`border`	`1px solid #ccc`
`width`, `max-width`	Size; `max-width` stops lines getting too long

The box model

From the inside out: **content → padding → border → margin**. Add `box-sizing: border-box;` so that `width` includes the padding and border.

Flexbox: a row of boxes

```
.row {
  display: flex;
  gap: 1rem;            /* space between items */
  justify-content: space-between;
  align-items: center;
  flex-wrap: wrap;      /* wrap on small screens */
}
```

> iPad tips: text in form fields at least 16px (`1rem`) so Safari does not zoom in; buttons and links at least 44px tall; and always keep the `viewport` meta tag.